

B. JOEL GONZALEZ

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EDUCATION

Carnegie Mellon University (CMU)

Pittsburgh, PA

College of Engineering

2021 - Present

- Ph.D. Candidate in *Electrical and Computer Engineering*
- M.S. in *Electrical and Computer Engineering*, concentration in *Integrated Systems* — May 2024
- Research Advisor: Dr. L. Richard Carley
- Research Interests: integrated systems design, nanofabrication, metrology and instrumentation

University of Pennsylvania (UPenn)

Philadelphia, PA

School of Engineering & Applied Science

2015 - 2021

- M.S.E. in *Robotics* — May 2021
- B.S.E. in *Computer Science*, minor in *Mathematics* — December 2019

TEACHING ASSISTANTSHIPS

Semesters as lead TA denoted by *.

CMU:

- 18-723: *RFIC Design and Implementation* — Spring 2026, Fall 2026
- 18-427/727: *Board-Level RF Systems for IoT* — Fall 2022*, Fall 2023*, Fall 2024*, Fall 2025*, Spring 2027*

UPenn:

- ESE350/ESE519: *Embedded Systems Lab* — Fall 2020, Spring 2021*
- CIS371/CIS501: *Computer Architecture* — Fall 2019, Spring 2020
- CIS380/CIS548/CIT595: *Operating Systems* — Fall 2019, Spring 2020*, Summer 2020*
- CIS240/CIT593: *Introduction to Computer Systems* — Fall 2018, Spring 2019, Summer 2019

WORKS

Authors contributed equally to the work denoted by *.

Publications and Preprints

- S. Hussain, B.J. Gonzalez, L. Chomas, and L.R. Carley. "Anisotropic high-aspect-ratio etching of sub-100-nm-diameter columns in silicon oxide." *MRS Advances* (2026). <https://doi.org/10.1557/s43580-026-01684-7>
- B.J. Gonzalez, E. Bourcart, J.K. Wirant, M. Juan, J. Wang, M.T. Moneck. "A low-cost, open-source maskless photolithography stepper for microfabrication". *arXiv preprint* (2025). <https://doi.org/10.48550/arXiv.2510.15082>

Presentations

- B.J. Gonzalez*, Y.I. Chen*, J.K. Wirant, J.S. Kunselman, E. Bourcart, M.T. Moneck, T. Srimani, and L. Pileggi (2026, May 27). "The Hacker Fab: An Open-Source Initiative for Nanofabrication Education" [Lecture]. ISCAS 2026, Shanghai, China.
- B.J. Gonzalez*, Y.I. Chen*, J.K. Wirant, J.S. Kunselman, E. Bourcart, M.T. Moneck, T. Srimani, and L. Pileggi (2026, May 26). "Live Demonstration: The Hacker Fab Maskless Photolithography Stepper" [Demonstration]. ISCAS 2026, Shanghai, China.
- S. Hussain, B.J. Gonzalez, L. Chomas, and L.R. Carley (2025, December 2). "Anisotropic High Aspect Ratio Etching of Sub-100nm Diameter Columns in Silicon Oxide" [Poster Presentation]. Materials Research Society Fall 2025 Meeting, Boston, MA, United States.
- B.J. Gonzalez, M.T. Moneck, and T. Srimani (2025, March 21-23). "The Hacker Fab: A DIY Approach to Nanofabrication" [Exhibition]. 2025 ECEDHA Annual Conference & ECEExpo, Norfolk, VA, United States.

EXPERIENCE

MIT Lincoln Laboratory

Lexington, MA

Division of Communication Systems

May 2023 - August 2023

- Summer 2023 research intern working on RF systems design and analysis for communication systems
- Programmed in Python and used RF hardware to analyze RF networks, spectra, and phase noise
- Internship completed as part of employer sponsorship via GEM Fellowship

ACTIVITIES, AWARDS, AND RECOGNITIONS

CMU:

- Carnegie Institute of Technology Dean's Fellow
- Recipient of the *GEM Fellowship*, with sponsorship from *MIT Lincoln Lab* for Summer 2023
- Completed the *Future Faculty Program* in Fall 2022 (Eberly Center for Teaching Excellence and Educational Innovation)
- Selected to participate in the 2022 *NextProf Pathfinder* workshop hosted by UMich and UCSD
- 2022-25 *CMU Robotics Club* Officer

UPenn:

- 2019-20 *CIS J.P. Eckert Fellow*
- 2020 inductee of the *CIS Max Mintz Undergraduate TA Hall of Fame*
- Honorable Mention for the 2021 Penn Engineering Outstanding Teaching Award
- Recipient of the *Summer 2020 TA Award for Excellence in Student Support with Distinction*
- Recipient of the 2019 *Littlejohn Scholars Summer Research Grant*
- Recipient of the 2018 *Penn Engineering Exceptional Service Award*
- Recipient of the 2017 *Penn Undergraduate Research Mentoring (PURM) Grant*